

SHOW ME YOUR AGENT / BUSINESS PROPOSAL

A purchasing plan that stays useful.

Kitchen Inventory Planning / Team 2B22 / NW0H3J8Y

ReStock helps a restaurant manager answer **what to buy, how much, when it can arrive, from whom, and whether the plan is still valid**. It combines recorded sales, inventory observations, expiry and supplier changes into evidence-backed purchasing recommendations. It supports decisions; it does not place supplier orders or make payments. [B1,B4]

The business proposition

Reduce avoidable purchasing mistakes and manual reconciliation by investigating changes before an outdated plan is acted upon. **AI organises the investigation; deterministic calculations and manager approval control the purchasing decision.**

Submission identity	Details
Team	2B22 / NW0H3J8Y
Problem statement	Kitchen Inventory Planning
Repository	github.com/rudybrrr/restock-ai
Public deployment	https://restock.13.250.80.71.sslip.io Organiser Lightsail. Authenticated manager session and a supported live organiser-gateway assessment were verified to REVISE_PLAN / ENGINE / PENDING_APPROVAL . Hosted exact-version approval, hosted safe escalation and restart persistence were not acceptance-tested.

01 / PROBLEM & OPPORTUNITY

One kitchen. Many changing assumptions.

The initial customer is a **manager-led small restaurant**, not an enterprise rollout. A plausible purchase list can become wrong by service time when demand changes, a supplier delivers less than expected, sales accelerate, or a physical count disagrees with the sales-based estimate. The manager must understand how those events change ingredient needs *and* commitments already made.

Operational friction	Business consequence	ReStock response
Sales and stock disagree	Purchasing against an inaccurate estimate; repeated reconciliation.	Separate physical observations from sales-based estimates and expose gaps.
Plan assumptions change	An obsolete recommendation may still be reviewed or approved.	Reassess material changes, retain history and reject stale approval.
Partial deliveries / shortages	Emergency buying can duplicate stock already received or still due.	Preserve fixed supply and propose only the additional uncovered quantity.
Perishability / constraints	A cheap offer may arrive too late or violate pack, storage or budget limits.	Check dated arrivals, expiry, MOQ/pack, storage, safety and the frozen budget.

PRODUCTIVITY Reduce repeated manual reconciliation and rework around changing plans.	COST Avoid duplicate emergency commitments; no measured savings are claimed yet.	RISK Exact-version approval, safe escalation and no automatic ordering/payment.	SCALE Start with one restaurant and limited menu; expand only after pilot evidence.
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Accountability and evidence basis. The restaurant manager owns buying and approval. Staff provide reliable sales intervals, receipts and lot-level counts; supplier terms are maintained by an authorised operator. This submission uses a synthetic five-dish, eight-ingredient restaurant with three suppliers. Customer needs are prototype hypotheses, not validated SME findings. Chains, automatic POS feeds and supplier checkout are later work. [B1,B4,B5]

02 / SOLUTION & BUSINESS VALUE

Fit the manager's day. Make the next action clear.

ReStock is more than a forecasting screen. Its value is **continuity of the decision**: observe a change, investigate the relevant evidence, keep or revise the plan when supported, and escalate when evidence or policy is insufficient. The result remains linked to captured inputs and the exact plan version so the manager can see what changed and what remains uncertain. [B1,B4]

Moment	Manager action	Intended value
During service	Report complete sales intervals; inspect stock estimates and stored forecasts/projections.	Spot changed demand and reporting gaps before acting on an obsolete plan.
Reviewing a plan	Compare quantities, supplier arrivals, cash and evidence; approve the exact version.	Reduce reconciliation while retaining human purchasing authority.
After arranging supply	Record the external purchase and partial/full receipts; report delays or shortfalls.	Avoid confusing proposed supply with actual commitments or arrived stock.
At closing	Submit final sales and per-lot physical counts; retain corrections and history.	Restore a trustworthy starting point for the next decision.

Adoption design - useful, not overclaimed Five task areas and local subtabs group related actions. A first-login spotlight tour can be skipped or replayed; detailed evidence stays expandable. These are usability features, not measured proof that onboarding time has fallen. The prototype has one configured manager role, not a complete staff-access product. A pilot also carries onboarding work: menu/recipe mapping, unit conventions, lot/expiry discipline and maintained supplier terms. [B4]	Demonstrated benefit mechanism - not an ROI claim In the bounded contingency fixture, 6 kg opening stock + 2 kg timely fixed supply = 8 kg against a 10 kg requirement. ReStock recommends only the missing 2 kg vegetables: S\$4 goods + S\$3 delivery + S\$4 emergency fee = S\$11 new purchase cash. This demonstrates avoiding duplicate commitments; it does not establish savings against a real restaurant baseline. [B5]
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Safe failure is part of the value. A completed assessment may escalate without a recommendation. Incomplete calculation, unknown data or an unsupported issue time must not be presented as sufficient stock or a zero-cost plan. If a model, calculation or required input is unavailable, the manager falls back to current operational facts and recorded reasons; unsupported automation stops rather than being disguised as success. [B4,B5]

03 / IMPACT & OUTCOMES

Measure useful decisions, not just more automation.

Three proposed v5 pilot targets; none is an achieved outcome. They test productivity, correctness and cost under matched conditions. [B1]

Proposed target	Definition / comparison	Evidence required
Planning effort 30% fewer manual interventions	Reduction = (baseline actions - ReStock actions) / baseline actions. Count review, correction and exception actions consistently in matched static/manual vs ReStock scenarios.	Instrument manager actions per scenario and actions per decision. If the baseline denominator is zero, report absolute values.
Decision correctness At least 90% correct keep/revise/escalate	Correct decisions / eligible disruption cases using pre-labelled expected actions; compare with a rule baseline on identical frozen inputs.	Held-out labelled cases; report missed and unnecessary replans and exclusions. The historical seven-case result is not this acceptance benchmark.
Operational cost At least 10% lower	(static cost - ReStock cost) / static cost over a matched simulated week, including purchase cash, waste valuation, stockout/lost-sales loss and emergency cost without double counting.	Approved valuation/terminal policies and complete histories. Full economics and observed-waste lifecycle remain disconnected, so this cannot yet be evaluated end to end.

Evidence discipline and guardrails Measure the baseline first; pre-register scenarios and exclusions; keep catalogue, initial stock, demand, supplier events and operating rules identical across comparators; publish scenario counts, denominators and failure cases. Do not improve a KPI by removing mandatory approval, bypassing stale-version rejection, increasing shortages/unmet demand under comparable inputs, or penalising a correct safe escalation. Monitor waste only after an authoritative recording lifecycle exists. Engineering tests and HTTPS reachability are evidence of implementation, not substitutes for business outcomes.

04 / FEASIBILITY & SCALABILITY

Start bounded. Earn the right to expand.

Stage	Acceptance gate	Owner
1. Submission prototype	Supported normal and contingency paths, safe escalation, recorded evidence and verified hosted normal run. Demo video and deployment evidence are separate submission items.	Ethan + Rudy; Backend support from Chun Yang
2. Shadow-mode pilot	One consenting restaurant; configure real recipes, units, suppliers and expiry conventions. Compare recommendations with existing practice without automatic buying.	Restaurant manager + implementation team
3. Broader evaluation	Freeze waste lifecycle and full-economic policies/contracts, then run matched and held-out benchmarks before claiming savings.	Chun Yang + Aniq; Rudy/Ethan integrate certified reads
4. Controlled adoption	Manager-approved use; train staff; monitor reporting completeness, exceptions, cost and recovery. Expand only if outcome and safety gates hold.	Restaurant manager + technical operator

Deployment, operations & commercial path

The organiser Lightsail lease showed a **US\$100 budget**; the selected instance was **US\$24/month, 4 GB RAM, 2 vCPUs, 80 GB SSD**. That is not an all-in operating quote: model usage, storage/backups, taxes, support and recovery costs still require measurement. No gateway unit price has been verified. [B6,B7]

Begin as an **assisted pilot**; consider a per-outlet service only after measuring manager value and operating cost. **No price, revenue, market-size estimate or paying customer is claimed.** The team owns implementation/reliability; the restaurant owns purchasing decisions and operational data quality.

Scalability path: prove one outlet first, then test more menus/users/transactions with monitoring, security, governance and cross-role access before any chain rollout.

Current boundaries & mitigation

- **Synthetic data:** validate recipes, reporting cadence, demand profiles and supplier terms in shadow mode; do not infer real impact from fixtures.
- **Bad/stale inputs or unsafe autonomy:** preserve unknowns and provenance; deterministic checks, exact-version approval and escalation; no automatic supplier ordering/payment.
- **Waste/full economics:** numerical helpers exist, but authoritative recording/reversal/replay and persisted certified economics remain incomplete; freeze policies before integration.
- **Hosted scope:** authenticated login and a supported live normal assessment are verified; hosted exact-version approval, safe escalation and restart persistence remain unverified.
- **Operational fallback:** when a model, calculation or input is unavailable, use current facts and manual review rather than inventing a recommendation.
- **Staff burden/KPI gaming:** track action counts and reporting completeness; fewer escalations are not inherently better.

05 / EVIDENCE & SUBMISSION

A credible proposal includes its boundaries.

Verified for submission: repository on main **eba56bf**; **1,341 backend PostgreSQL tests passed** on the recording commit; Ruff clean; Pyright 0 errors/warnings; frontend lint and production build passed; authenticated Lightsail manager session and a supported live organiser-gateway assessment completed to **REVISE_PLAN / ENGINE / PENDING_APPROVAL**. The final demo also shows bounded contingency, local material-sales safe escalation and stale-version protection. [B5,B8]

Not claimed: real restaurant impact, measured savings, provider reliability, automatic POS integration, supplier checkout, connected observed-waste lifecycle, actionable full 21-day economics, production support cost, market size or proven willingness to pay.

Proposal-to-demo consistency

Claim: the plan stays valid as reality changes. **Demo evidence:** a supported normal assessment, exact-version human control, a deterministic post-purchase shortfall that recommends only the additional 2 kg / S\$11, and safe escalation when normal purchasing is no longer supported.

Source register. [B1] *ReStock_AI_Hackathon_Plan_v5_Data_Cadence_Clarified.md* - product intent and proposed 30% / 90% / 10% targets. [B2] *ReStock ML Decision Engine Plan v2* - numerical design and open policy questions. [B3] *ML v3 Integration Amendment* - first-slice alignment and cash/full-economics scope. [B4] *docs/ARCHITECTURE.md* - authority/workflow and product boundaries. [B5] *docs/FINAL_VALIDATION_2026-09-26.md + POST_PURCHASE_CONTINGENCY_CONTRACT.md* - dated scenarios and exact contingency. [B6] latest organiser submission instructions - separate PDFs, demo video and deployment evidence; organiser Lightsail live URL required. [B7] 27 Sep Lightsail operator evidence - selected instance, four services active, HTTPS /login 200. [B8] final demo recording/QA, 27-28 Sep - main **eba56bf** final test/build gate, authenticated hosted manager session and supported live normal assessment.

Submission package

Team code and problem statement in the email; GitHub repository URL; this Business Proposal PDF; Technical Document PDF; demo video URL; and Lightsail deployment evidence URL/artifact. Email subject: **SMYA Final Submission - NW0H3J8Y**.